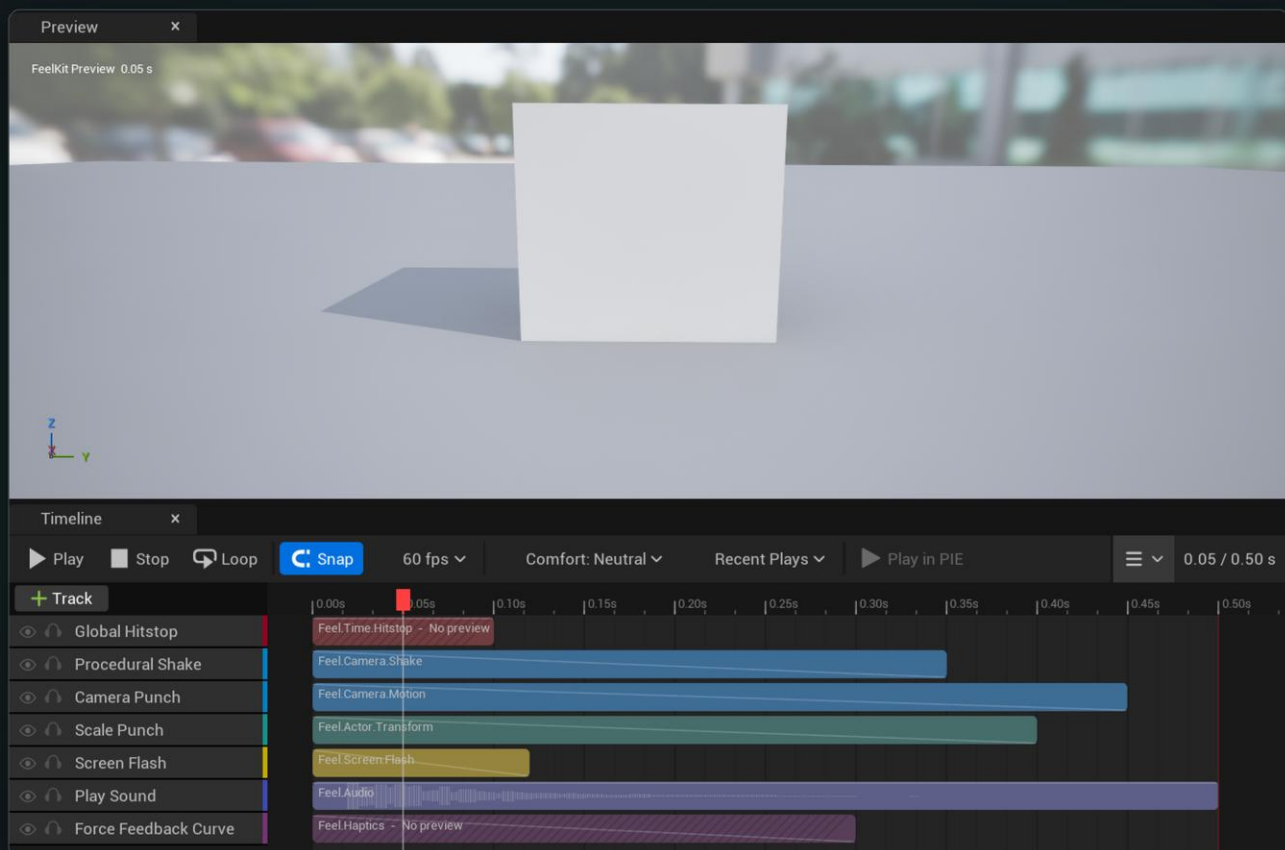


UNREAL ENGINE PLUGIN

FeelKit

Manual

Build game feel on a timeline. Play it with one node.
Let every player choose how much they feel.



TIMELINE

Build and preview a hit without pressing Play.

ONE NODE

Play recipes from Blueprint or C++, or send an event.

COMFORT

Every player chooses how much shake, flash and rumble they get.

Contents

- 2 Core ideas.....4
 - 2.1 Recipes, tracks and steps4
 - 2.2 Channels, and what happens when effects overlap5
- 16 Step reference8
 - 16.5 Camera steps8

PRO ONLY In FeelKit Pro only. These sections also carry an orange strip under the heading.

LITE + PRO In both editions. Guide sections without a tag are in both editions too.

Draft, 24 September 2026

PART ONE

Using FeelKit

- 1 Welcome
- 2 Core ideas
- 3 Installing FeelKit
- 4 Quick start: your first recipe
- 5 The recipe editor
- 6 Playing recipes in your game
- 7 Parameters, context and variation
- 8 Comfort and accessibility
- 9 The recipe library and browser
- 10 Debugging and tuning
- 11 Multiplayer
- 12 Performance
- 13 Demo kits
- 14 Troubleshooting and questions
- 15 Lite and Pro

2 Core ideas

IN THIS CHAPTER

- 2.1 Recipes, tracks and steps
- 2.2 Channels, and what happens when effects overlap

2.1 Recipes, tracks and steps

A **recipe** is an asset that describes how one moment of the game feels: a hit landing, a heavy landing after a fall, a pickup, a menu button being pressed. It holds **tracks** laid out on a timeline, the way a Level Sequence holds tracks in Sequencer. Each track plays one **step**, which is the effect itself: a camera shake, a screen flash, a sound, a controller vibration, a short freeze of time. FeelKit Pro has 37 steps and FeelKit Lite has 12 of them; [chapter 16, Step reference](#) describes each step and names its edition.

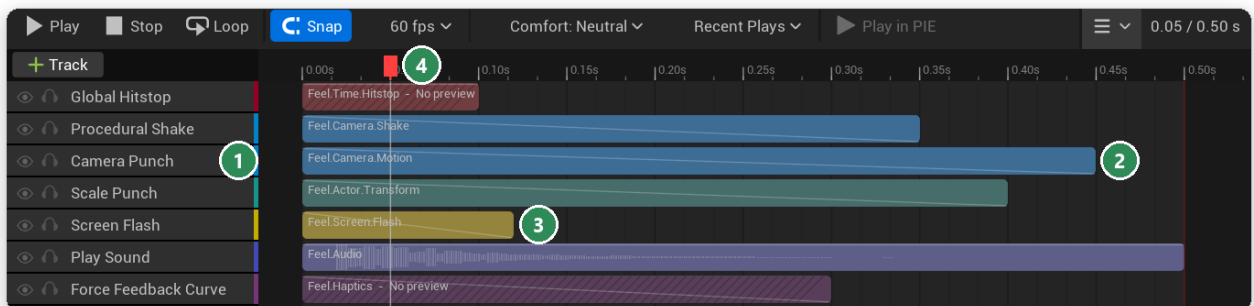


Figure 2.1 The recipe editor on a copy of FR_Impact_HeavyHit. 1 the steps, one per track; 2 a track on the timeline; 3 its channel, by name and colour; 4 the playhead

When the game plays a recipe, every track starts at its **Start Time** and plays its step for its **Duration**, both in seconds. A new track lasts 0.5 s. A track with a length of 0 is an instant track: steps that act once, such as Play Sound, Spawn Decal or Blueprint Event, can sit on one. The play ends when its last track ends.

A few settings belong to the recipe as a whole:

- **Cooldown** (default 0 s): how long the same recipe must wait before it can play again on the same target.
- **Max Concurrent** (default 0, which means no limit): how many plays of the recipe can run at once on one target. When the limit is reached, a new play does not start and the running ones continue.

- **Default Intensity** (default 1): a multiplier on every play of the recipe.

The full list of recipe and track fields is in [16.1 Recipe fields](#) and [16.2 Track fields](#).

A step works out its output for a moment from three things only: the time inside its track, its intensity at that time, and a seed. It keeps nothing from earlier frames. That is why the recipe editor can scrub the timeline forwards and backwards and show exactly the frame the game would show at that moment, and why the same seed always gives the same shake. Steps that act once, such as playing a sound or calling a Blueprint event, do so when their track starts, and some act again when it stops.

Recipes are timed in real time, which Unreal calls unscaled time. A hitstop that slows the world does not slow the recipe that caused it: a shake keeps moving during the freeze, and the recipe ends when its timeline says. FeelKit's runtime only does work while a recipe plays or while something it changed still has to be put back; with nothing playing, it does not tick.

2.2 Channels, and what happens when effects overlap

Every track has a **channel**, a gameplay tag under `Feel`. A new track takes its step's default channel, for example `Feel.Camera.Motion` for Camera Punch, and you can change it on the track. The channel colours the track on the timeline, groups the recipe editor's **Intensity** tab, and decides which comfort group scales the track (see [2.7 Comfort on one page](#)).

Channel	Used by	Comfort group
<code>Feel.Camera.Shake</code>	Procedural Shake	Camera Shake
<code>Feel.Camera.Motion</code>	Camera Punch, FOV Kick, Camera Roll, Camera Zoom, Look-at Nudge	Camera Motion
<code>Feel.Time.Hitstop</code>	Global Hitstop, Actor Hitstop	Hitstop and Slow-mo
<code>Feel.Time.SlowMo</code>	Slow-mo Ramp	Hitstop and Slow-mo
<code>Feel.Screen.Flash</code>	Screen Flash	Flashes
<code>Feel.Screen.Distortion</code>	Vignette Pulse, Chromatic Aberration, Post Process Material Pulse	Screen Distortion
<code>Feel.Screen.Color</code>	Desaturate, Color Tint	Screen Distortion
<code>Feel.Screen.Fade</code>	Screen Fade	none (Master only)
<code>Feel.Actor.Transform</code>	Scale Punch, Squash and Stretch, Mesh Wobble	none (Master only)
<code>Feel.Actor.Material</code>	Material Parameter Pulse, Hit Flash	none (Master only)
<code>Feel.Actor.Light</code>	Light Flash	Flashes
<code>Feel.Audio</code>	Play Sound	none (Master only)
<code>Feel.Audio.Mix</code>	Sound Class Duck, Pitch Bend, Low-pass Sweep	none (Master only)
<code>Feel.Haptics</code>	Force Feedback Curve, Haptic Pattern	Haptics
<code>Feel.UI</code>	Widget Punch, Widget Shake, Widget Flash, Number Pop	none (Master only)

Channel	Used by	Comfort group
Feel.Spawn	Spawn Decal, Spawn Particle	none (Master only)
Feel.Meta.Event	Blueprint Event	none (Master only)
Feel.Meta.Recipe	Play Recipe, Random Choice	none (Master only)

The comfort groups in this table are the project defaults. You can change the mapping in the project settings (see [18.5 Comfort](#)).

PART TWO

Reference

- 16 Step reference
- 17 Blueprint nodes and components
- 18 Project settings and editor preferences
- 19 Console commands
- 20 Writing your own steps in Blueprint and C++
- 21 Recipe JSON files and editor scripting
- 22 Credits and licences
- 23 Version history

16 Step reference

16.5 Camera steps

Camera steps offset the view of the target's local player, through a camera modifier that FeelKit adds to each local player's camera manager. Tracks of one play add up; separate plays are combined by the **Camera Arbitration** setting (see [18.2 Camera](#)).

16.5.1 Procedural Shake LITE + PRO

Moves and turns the camera with noise or a sine wave computed from the track time and the seed, so the same seed gives the same shake every time and the preview shows exact frames when scrubbing. **Perlin** gives smooth random motion on each axis. **Sine** gives a regular oscillation with a random phase per axis. **Directional** moves back and forth along **Direction** only, with the rotation swinging in step. The intensity curve sets the strength over time; the default curve gives a shake that dies away over the track.

Class	UFeelStep_ProceduralShake, module FeelCore.
Default channel	Feel.Camera.Shake.
Comfort group	Camera Shake.
Editor preview	yes.
Track length	needs a length.
New tracks start with	an intensity curve that fades from 1 to 0.

Property	Default	Description
Mode	Perlin	Shape of the shake.
Frequency	12 Hz	Oscillations per second. At least 0.01.
Location Amplitude	(0, 3, 3) cm	Maximum location offset per camera axis (X forward, Y right, Z up). Shown only when Mode is not Directional .
Direction	(0, 0, 1)	Camera-space direction of a directional shake. Shown only when Mode is Directional .
Directional Amplitude	8 cm	Maximum offset along Direction. Shown only when Mode is Directional .
Rotation Amplitude	Pitch 1, Yaw 1, Roll 0.5 (degrees)	Maximum rotation offset in degrees per axis.
Field Of View Amplitude	0°	Maximum field of view change.

16.5.2 Camera Punch LITE + PRO

Pushes and turns the camera to a peak and back along the motion shape; with the default Spring it overshoots and settles like a physical hit. **Location Punch** and **Rotation Punch** are the offsets at the peak. **Direction Source** can point the punch along the play's Direction, or away from or toward the play's Location, as seen from the camera when the play started. The strength stays that of the step settings, and the step settings are used when the play passed no direction or location. **Direction Jitter** turns the direction by a random angle for each play, which stays fixed for the whole play.

Class	UFeelStep_CameraPunch, module FeelCore.
Default channel	Feel.Camera.Motion.
Comfort group	Camera Motion.
Editor preview	yes.
Track length	needs a length.
New tracks start with	a flat intensity curve at 1.
Motion shape	Spring by default; the shape settings are described in 16.4 Shared motion settings .

Property	Default	Description
Location Punch	(-10, 0, -4) cm	Camera-space location offset at the peak (X forward, Y right, Z up).
Rotation Punch	Pitch -2.5, Yaw 0, Roll 0 (degrees)	Rotation offset at the peak, in degrees.
Direction Source	Step Settings	Where the punch direction comes from. With a play direction or location, the punch keeps the strength of Location Punch and Rotation Punch but points along that direction, as seen from the camera when the play started.
Direction Jitter	0°	Varies the punch direction per play by up to this angle, so repeated plays never look identical. Range 0 to 90.

16.5.3 FOV Kick LITE + PRO

Changes the field of view by **Field Of View Kick** at the peak and back. Positive values widen the view, which reads as speed; negative values narrow it, which reads as an impact zoom. The default Kick shape rises fast and eases back.

Class	UFeelStep_FOVKick, module FeelCore.
Default channel	Feel.Camera.Motion.
Comfort group	Camera Motion.
Editor preview	yes.
Track length	needs a length.
New tracks start with	a flat intensity curve at 1.
Motion shape	Kick by default; the shape settings are described in 16.4 Shared motion settings .

Property	Default	Description
Field Of View Kick	8°	Field of view change at the peak. Positive widens (speed), negative narrows (impact zoom). Range -60 to 60.

16.5.4 Camera Roll PRO ONLY

FeelKit Pro only. FeelKit Lite does not include this.

Rolls the camera around its view axis to **Roll Degrees** at the peak and back. With **Random Direction** on, each play picks clockwise or counterclockwise from its seed. Players who turn **Allow Camera Roll** off in their comfort settings see no camera roll from any recipe.

Class	UFeelStep_CameraRoll, module FeelCore.
Default channel	Feel.Camera.Motion.
Comfort group	Camera Motion; the roll is also removed completely when a player turns Allow Camera Roll off.
Editor preview	yes.
Track length	needs a length.
New tracks start with	a flat intensity curve at 1.
Motion shape	Spring by default; the shape settings are described in 16.4 Shared motion settings .

Property	Default	Description
Roll Degrees	4°	Roll angle at the peak. Positive rolls clockwise. Range -45 to 45.
Random Direction	On	Pick clockwise or counterclockwise at random for each play.